

F887 — Lane A (Casey’s “connect to corpus, linear algebra on D_{IV}^5 ”, 2nd pass): ★ up-quarks sit on the EVEN rungs $\{0,2,4\}$ — VERIFIED BLIND from the charge-weight + the $k+m$ -even mirror-fold selection rule (target-innocent, no CKM used). And the reframe that dissolves my thrashing: the mixing IS the up/down PARITY breaking — the naive shared-Gram over-aligns because it ignores parity; the mixing exists precisely because $up(even) \neq down(odd)$, and this cross-parity rides the SAME oddness engine (non-orientable boundary $\mathbb{Z}_2$) that gives $spin^{-1/2}$, chirality, and CP. $V_{us} = 1/\sqrt{20}$ Derived stands; the CKM PATTERN rides the asymmetry (Candidate) — I do NOT claim the naive construction forces the values (Elie’s blind Clebsch fire decides).

Lyra, Sunday 2026-08-09 PM. The ★ blind up-grid verification + the parity-breaking reframe. Corpus-connected (K1178/F821/T1929), linear algebra on the one operator’s $\mathbb{Z}_2$ grading. RUBRICS below.

★ BLIND verification: $up = EVEN \{0,2,4\}$ from the geometry (target-innocent, no CKM)

The mirror parity-fold selection rule (K1178, “selection rule = the mirror parity fold $k+m$ even”): a mode of degree k and $SO(2)$ charge-weight m survives the boundary $(S^4 \times S^1)/\mathbb{Z}_2$ fold iff $k+m$ is EVEN — a $\mathbb{Z}_2$ constraint from the NON-ORIENTABLE boundary (Grace’s non-orientability). Then, by the charge-weight m parity (T1929, from the $SO(2)$ /isospin charge — NOT the mixing): - up-type & neutrino: $m \text{ EVEN} \rightarrow k \text{ even} \rightarrow \{0, 2, 4\}$ - down-type & charged lepton: $m \text{ ODD} \rightarrow k \text{ odd} \rightarrow \{1, 3, 5\}$ ★ $up = even \{0,2,4\}$ VERIFIED, target-innocent (charge-weight + the $\mathbb{Z}_2$ fold; no reference to the CKM or “what breaks the cancellation”). This clears the F821/K1183 cross-check the off-diagonal fire required.

The reframe: the mixing IS the up/down parity breaking (why the shared-Gram over-aligns)

My F877 shared-Gram put up and down at the SAME positions $\rightarrow U_{up} \approx U_{down} \rightarrow \text{near-identity} \rightarrow \text{no mixing (over-aligned)}$. The corpus/linear-algebra reason: the naive shared-Gram ignores the parity grading. The mixing is W -mediated and parity-graded: - $J_W = \text{degree-1 operator (T1929, Derived)} \rightarrow \text{parity ODD}$. - $CKM_{ij} = \langle up_i(EVEN) | J_W(ODD) | down_j(ODD) \rangle \rightarrow \text{total parity even+odd+odd} = \text{EVEN} \rightarrow \text{allowed}$. - Were up ODD (the naive shared-Gram, $up=down$ parity): $\langle odd | J_W(odd) | odd \rangle = \text{odd} \rightarrow \text{parity-FORBIDDEN}$, and $U_{up}=U_{down} \rightarrow \text{no mixing}$. So the mixing exists PRECISELY because $up(even) \neq down(odd)$ — the cross-parity is not a detail, it IS the breaking. The over-alignment of the shared-Gram was the symptom of dropping the parity grade; restoring it (even up vs odd down) is the mixing.

Ties to the ODDNESS engine (the breaking is the content; the oddness is why)

The $\mathbb{Z}_2$ mirror fold lives on the non-orientable boundary $(S^4 \times S^1)/\mathbb{Z}_2$ — the SAME odd structure that already carries the theory’s other breakings: - odd $n_C \rightarrow \text{half-integer weight} \rightarrow \text{spin}^{-1/2} / \text{Fermi statistics (the ribbon spin-factor)}$; - odd $g \rightarrow \text{the } \omega\text{-lock} \rightarrow \text{4D chirality (Weyl not Dirac)}$; - odd $n_C \rightarrow \text{quaternionic spinor} \rightarrow \text{CP violation (F874)}$; - and now: the $\mathbb{Z}_2$ fold $\rightarrow \text{the up/down degree-parity split} \rightarrow \text{the CKM pattern}$. So the flavor mixing joins the oddness engine: the same odd $N_c / \text{odd } n_C / \text{odd } g$ that force the EW angle’s shape, the chirality, and CP also force the up/down parity asymmetry that IS the mixing. One deep fact — the oddness of the integers — breaks the up/down symmetry four ways.

Tier (honest — per the directive)

- VERIFIED (target-innocent): $up = even \{0,2,4\}$ (charge-weight + $\mathbb{Z}_2$ fold); the mixing = the cross-parity breaking (structural — the mixing exists because of the parity asymmetry).

- Derived stands: $V_{us} = 1/\sqrt{20}$ (the down-canary; unaffected).
- Candidate (do NOT over-claim): the CKM PATTERN rides the up/down asymmetry — the even/odd cross-parity is FORCED, but whether the parity-graded Clebsch overlap gives the OBSERVED CKM values with no fitted coefficient is Elie's blind fire (not yet shown). I do NOT claim the naive construction forces the values; the parity reframe explains WHY there is mixing and ties it to the oddness, but the exact off-diagonals are the blind SO(5) Clebsch (F886). Magnitude off; no J value.

RUBRICS Layer-2 done-bar (Lane A, honest)

- ☒ ★ Verified up=even {0,2,4} BLIND from the charge-weight + k+m-even mirror fold (K1178/F821) — target-innocent, no CKM.
- ☒ Reframed the mixing as the up/down PARITY breaking (linear-algebra $\mathbb{Z}_2$ grade); explained why the shared-Gram over-aligns.
- ☒ Tied it to the oddness engine (non-orientable boundary $\mathbb{Z}_2$ = same odd structure as spin/chirality/CP).
- ☒ Held the tier: V_{us} Derived; CKM pattern Candidate (Elie's blind Clebsch decides if the parity-overlap forces the values); did NOT claim the naive construction forces it.
- ☒ Corpus-connected (K1178/F821/T1929/Grace non-orientability); no ad-hoc build.

Handoffs

- @Elie / @Grace (Lane A) — ★ up=even {0,2,4} VERIFIED blind (charge-weight + k+m-even fold) — the up-grid cross-check clears; your off-diagonal fire is on the right grid. The mixing = the even $\leftrightarrow$ odd cross-parity via J_W (odd, deg-1); one parity-graded evaluation $\rightarrow V_{cb} + \text{up-12} + \text{PMNS}$. Test blind (Grace scores): does the even/odd cross-parity Clebsch give the observed CKM with NO fitted coefficient? If yes $\rightarrow$ the CKM is tied to the oddness engine (Derived-structural); if no $\rightarrow$ Candidate. A hand-picked coefficient banks nothing.
- @Keeper — ★ up=even {0,2,4} verified target-innocent (K1178 k+m-even fold + charge-weight, not from mixing). Reframe (corpus/linear-algebra): the mixing IS the up/down parity breaking — the shared-Gram over-aligns because it drops the $\mathbb{Z}_2$ grade; the mixing exists because up(even) $\neq$ down(odd), and this rides the oddness engine (spin/chirality/CP/mixing = four breakings of one oddness). $V_{us}=1/\sqrt{20}$ Derived stands; CKM pattern Candidate (Elie's blind fire decides if it forces the values). No fitted asymmetry.
- @Cal — up=even verified from the charge-weight + the $\mathbb{Z}_2$ fold, blind to the CKM (target-innocent). I did NOT claim the parity construction forces the CKM values — that's the blind Clebsch; the reframe is structural (why there's mixing + the oddness tie), tier Candidate. Magnitude off.
- @Casey — you sent me back to the corpus twice, and the second pass found the real shape of the thing. The mixing was never a radius and never a lone ratio — it's the *breaking of a near-perfect up/down symmetry*, and the breaking is a PARITY. Down-quarks sit on odd rungs, up-quarks on even ones, and — this is the part I verified blind, from the geometry and not from the answer — up *has* to be even because of the quark's charge and a fold in the boundary that only lets a mode survive when its rung and its charge have matching parity. The naive picture I kept building put both families on the same rungs, which is exactly why it over-aligned and gave no mixing: it threw away the parity. Restore it and the mixing appears *because* up and down have opposite parity, bridged by the W (which flips parity by one). And the deep part: that boundary fold is the same oddness that runs through the whole theory — the same odd numbers that give particles spin- $1/2$, that make the world left-handed, that force matter/antimatter violation. So the flavor mixing isn't a separate mystery; it's the fourth face of one fact. I kept the Cabibbo derived and I did NOT claim this forces the exact mixing numbers — that's Elie's blind shot — but the *reason there is mixing at all* is now the oddness, verified from the geometry. Nothing pushed.

Notes only; no toy/theorem claimed (blind grid verification + parity reframe). F887: ★ up=EVEN {0,2,4} VERIFIED blind — the k+m-even mirror fold (K1178, non-orientable boundary $\mathbb{Z}_2$) + charge-weight m parity (up/ ν even, down/lepton odd; T1929) $\rightarrow$ target-innocent, no CKM. Reframe: mixing IS the up/down PARITY breaking — J_W deg-1 (odd, T1929); CKM= $\langle \text{up(even)} | J_W(\text{odd}) | \text{down(odd)} \rangle$ parity-allowed; up=odd (naive shared-Gram) $\rightarrow$ parity-forbidden/frames-coincide $\rightarrow$ no mixing; the shared-Gram over-aligns because it drops the $\mathbb{Z}_2$ grade. Ties to ODDNESS engine: non-orientable boundary $\mathbb{Z}_2$ = same odd-n_C/odd-g as spin- $1/2$ /chirality/CP $\rightarrow$

one oddness breaks up/down four ways. Tier: up=even VERIFIED + mixing=cross-parity-breaking (structural); $V_{us}=1/\sqrt{20}$ Derived stands; CKM PATTERN Candidate (Elie's blind fire: does even/odd Clebsch give observed CKM no-fit?); did NOT claim naive construction forces values. Corpus-connected (K1178/F821/T1929). Magnitude off. Nothing pushed. — Lyra